Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is **a possible direct DoS SYN flood attack where they simulate the TCP connection and flood the server with SYN packets. The timeout error message is happening because of the web server not being able to handle the large number of SYN requests and not responding properly.** The logs show that **initially the web server is handling the requests well and we can recognize TCP connections from IP known in the company environment. Then an unfamiliar IP starts sending more and more SYN requests until the web server becomes unable to handle everything and starts timing out the requests.** This event could be **related to the DoS SYN flood attack as the threat actor is trying to overload SYN requests to our web server in order to overwhelm it and make it unable to function.**

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1. **The first step is to send a SYN request, or synchronize, to the web server. The employees of our company do this every time with their devices when they need to connect to a webpage and the request is sent to the webserver.** 2. **The web server gives the permission, responding with a SYN/ACK packet, to say they "synchronize acknowledge" the request and they find a port open for the final step.** 3.**The device resends the ACK received to the web server and the TCP connection is established, so in this situation, the connection to the webpage.** Explain what happens when a malicious actor sends a large number of SYN packets all at once: **So, when a SYN flood attack occurs, the threat actor is trying to take advantage of this protocol by flooding the web server with SYN packet requests in order to overwhelm the web server and occupy all its ports. This will lead to the web server not being able to process more requests and being unable to function properly.** Explain what the logs indicate and how that affects the server: **In this situation, the logs indicate that the IP address of the threat actor is the "203.0.113.0", which has infiltrated slowly sending some requests while the web server**

was working well and was managing the employees requests too. Then he started flooding the web server with SYN requests, creating a situation of slowness of responses from the web server, who was sending time out messages to the employees requests as it was unable to function properly, then only the SYN flood requests from the threat actor were passing through to the web server, leading it to be overwhelmed and unable to function.
An imminent solution should be to ban with the firewall this specific IP address from the threat actor, as it's a DoS attack, happening for now only from that specific IP.
To prevent this type of attack in the future can be done by adding a proxy server or a filter in order to recognize only the device IP address of the company network sending requests to the web server, and protecting it from external threat sending requests.